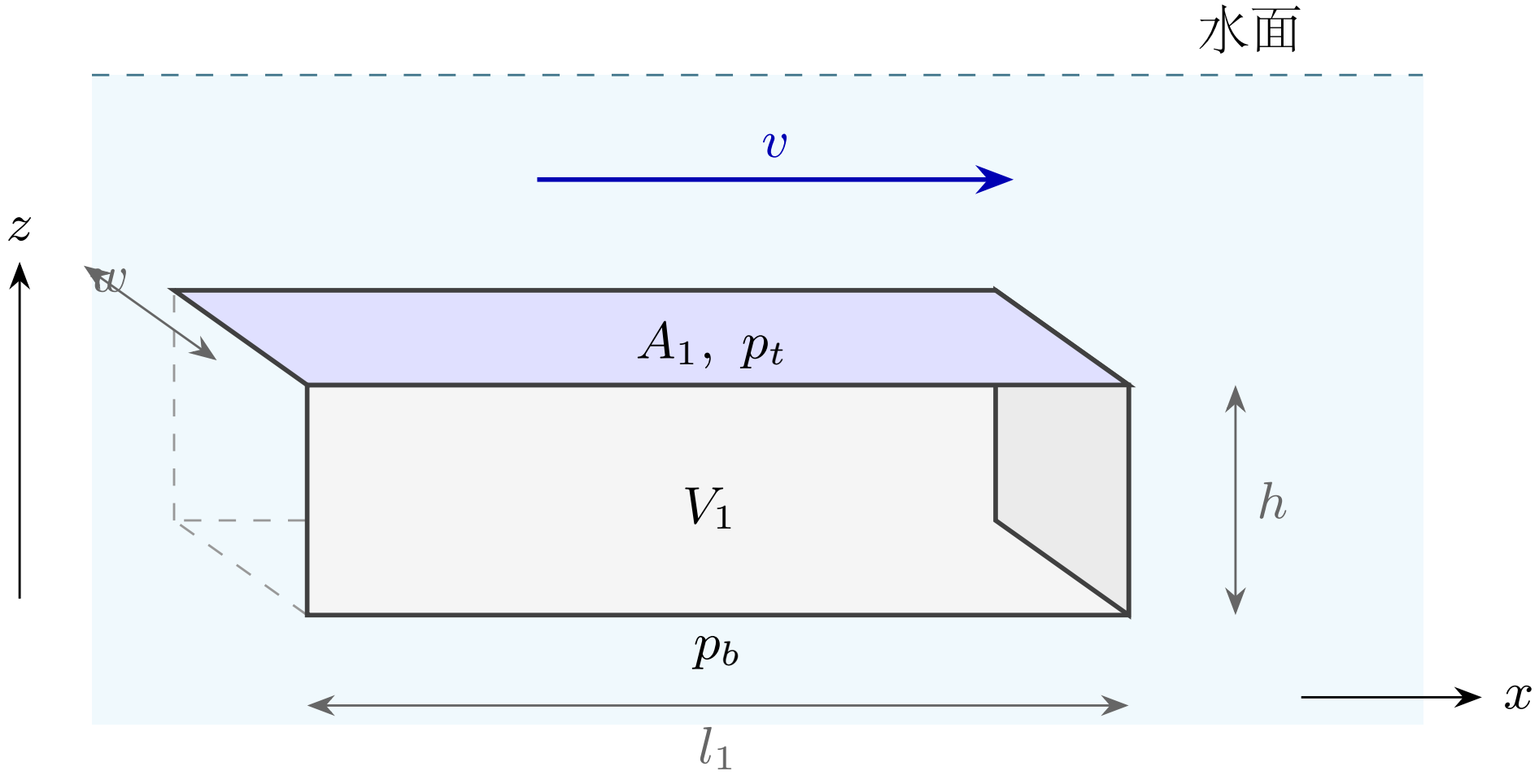
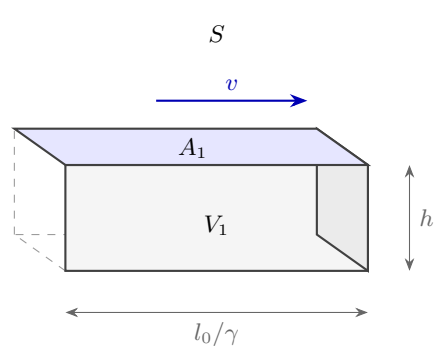
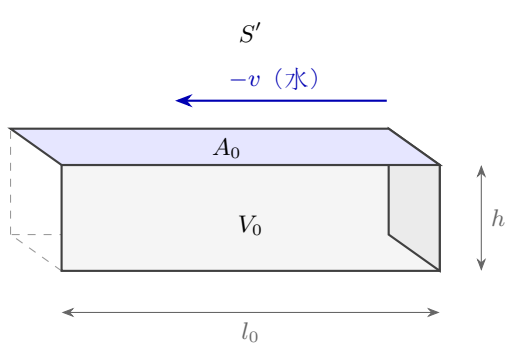
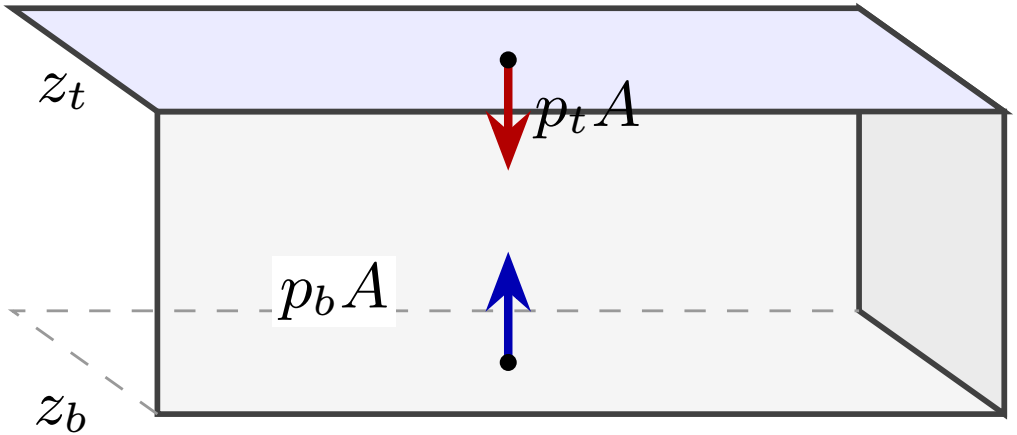

水面
v
z
w
$A_1,\ p_t$
V_1
h
p_b
x
l_1

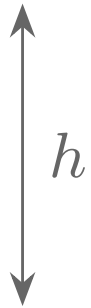


水静止，艇运动



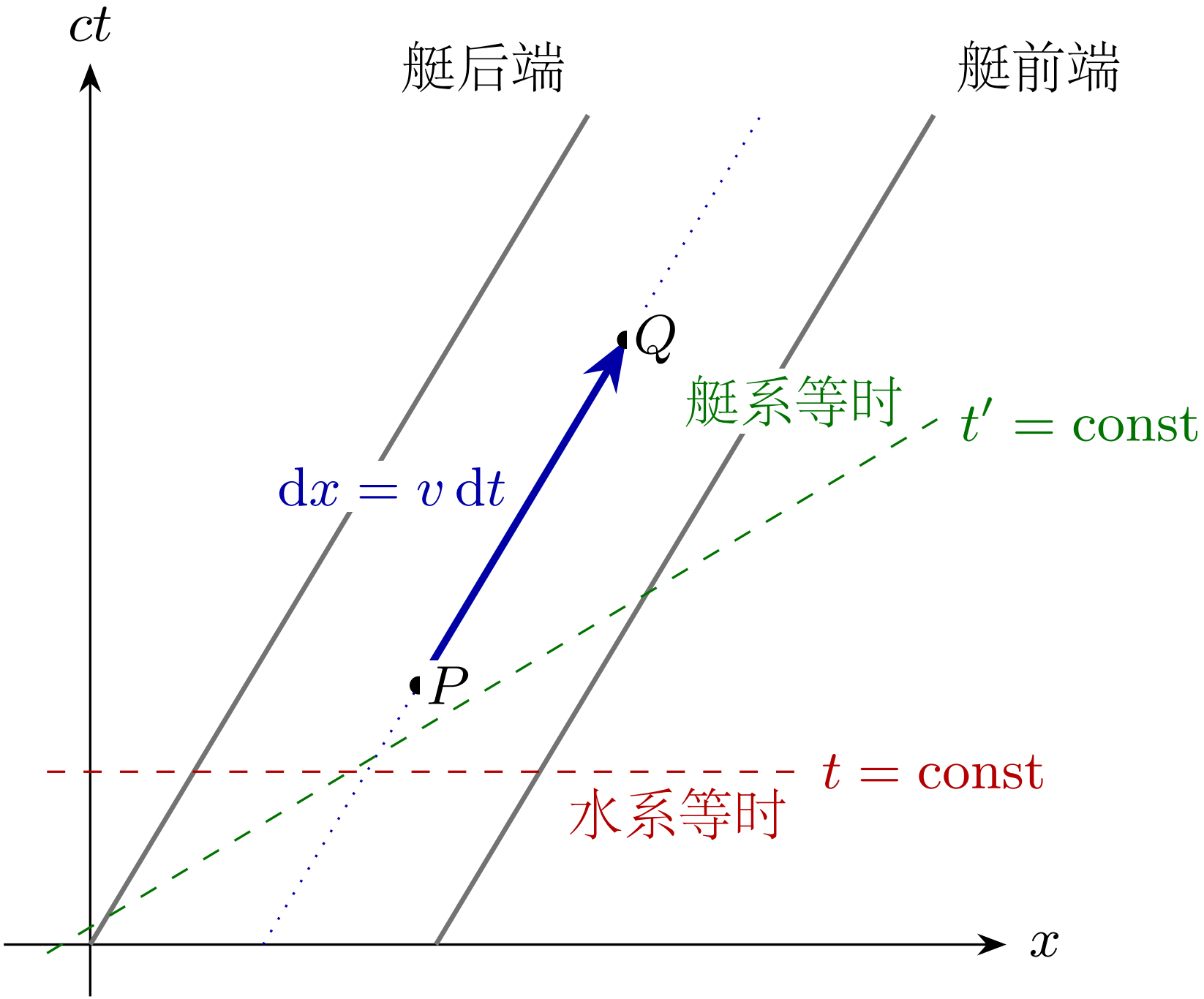
艇静止，水运动



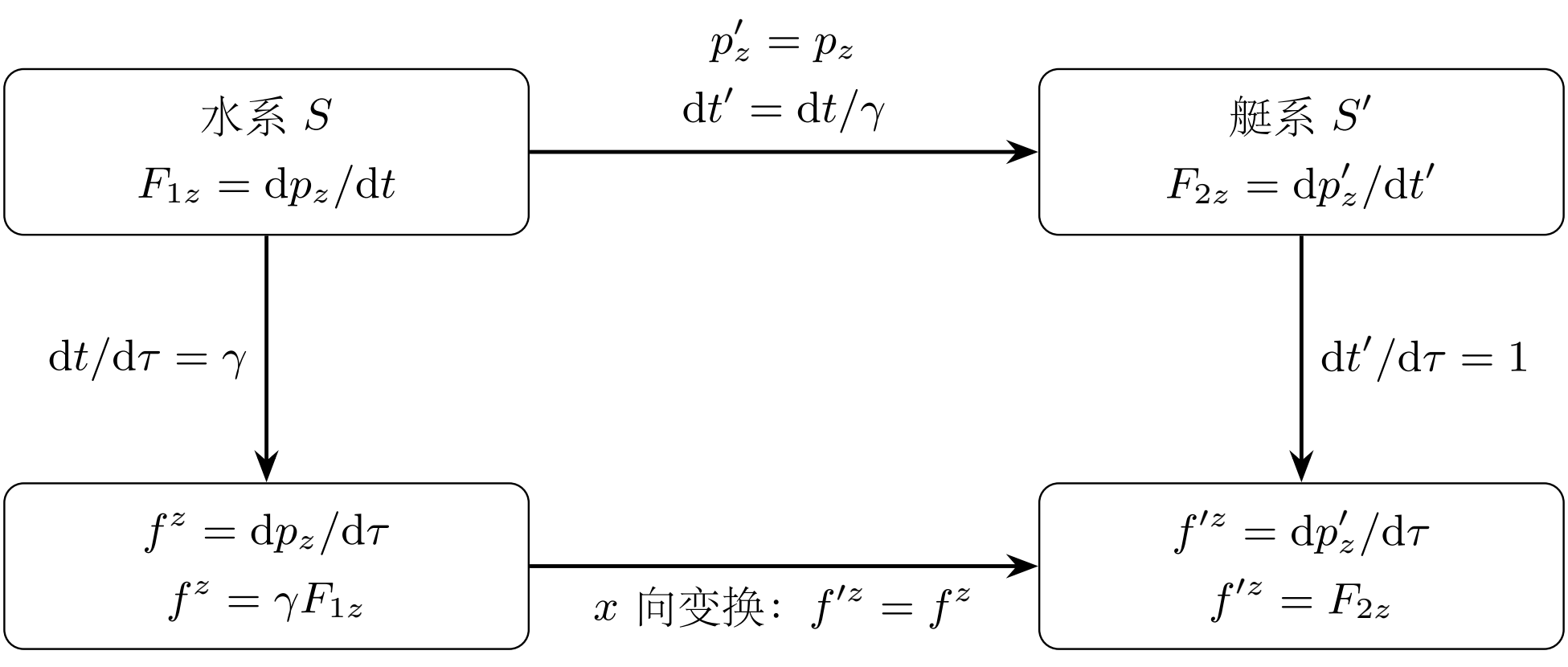


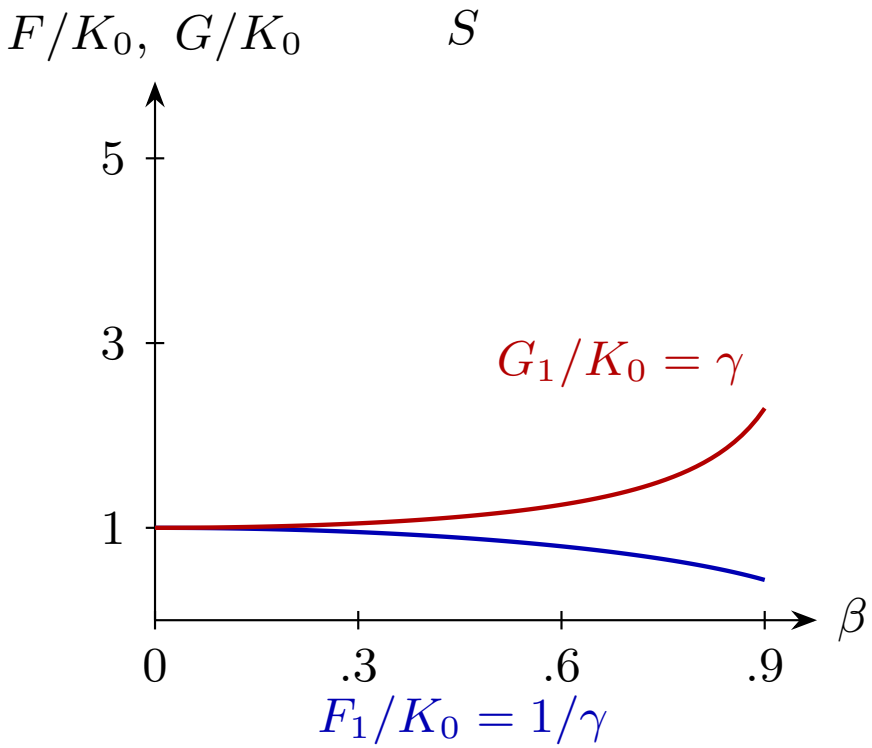
$$F = (p_b - p_t)A$$

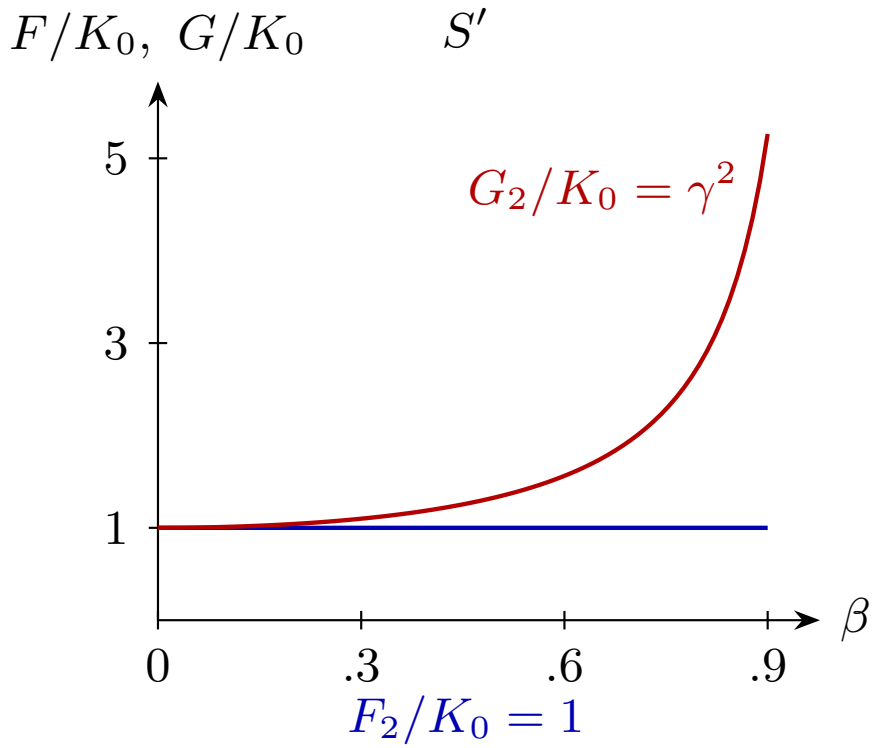
$$p_b - p_t = \rho_0 g h, \quad A = A_1 \ (S), \ A_0 \ (S')$$



P, Q 位于同一艇体质心世界线


水系 S
$F_{1z} = \mathrm{d}p_z/\mathrm{d}t$
$p'_z = p_z$
$\mathrm{d}t' = \mathrm{d}t/\gamma$
艇系 S'
$F_{2z} = \mathrm{d}p'_z/\mathrm{d}t'$
$\mathrm{d}t/\mathrm{d}\tau = \gamma$
$\mathrm{d}t'/\mathrm{d}\tau = 1$
$f^z = \mathrm{d}p_z/\mathrm{d}\tau$
$f^z = \gamma F_{1z}$
x 向变换：$f'^z = f^z$
$f'^z = \mathrm{d}p'_z/\mathrm{d}\tau$
$f'^z = F_{2z}$





$K_0=\rho_0 V_0 g$；两图的 β 为同一相对速度